

Middle Convolution of ℓ -adic Perverse Sheaves

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Contents

1	Introduction	2
1.1	Notations	2
1.2	Motivations	3
1.3	Structure of the Write-up	3
1.4	Acknowledgements	4
2	Local Systems and Lisse Sheaves	4
3	Perverse Sheaves and the Six Functor Formalism	4
4	Middle Convolution and Monodromy Groups	4
5	Mod-ℓ Monodromy of Hyperelliptic Curves	4

1 Introduction

1.1 Notations

1.1.1. Motivated by the growing interest in the formalization of mathematics, we adopt notation inspired by type theory throughout this write-up, without committing to type-theoretic foundations (foundational issues are irrelevant to this write-up). The symbols below denote types or categories depending on your preference, where $X : T$ denotes that X is a term of type T , or an object of the category T .

Set	Sets
Set^{fin}	Finite sets
Nat	Natural numbers (including zero)
Int	Integers
Ab	Abelian groups
Ring	Rings (associative, unital, not necessarily commutative)
Field	Fields
Mod_R	Left modules over $R : \text{Ring}$
Vect_k	Vector spaces over $k : \text{Field}$
$\text{Vect}_k^{\text{fin}}$	Finite-dimensional vector spaces over $k : \text{Field}$
Sch	Schemes
Sch_S	Schemes over base scheme $S : \text{Sch}$
Var_k	Varieties over $k : \text{Field}$, i.e. integral separated schemes of finite type over k
Curv_k	Curves over $k : \text{Field}$, i.e. one-dimensional varieties over k
$\text{Curv}_k^{\text{sm}}$	Smooth curves over $k : \text{Field}$
Riem	Riemann surfaces, i.e. connected one-dimensional complex manifolds
Site	Sites, i.e. categories equipped with Grothendieck topologies
$\text{Sh}(X, A)$	Sheaves on $X : \text{Site}$ with values in a category A admitting small limits
$\text{Mod}(X, \mathcal{R})$	Sheaves of left $\mathcal{R}$ -modules on $X : \text{Site}$, where $\mathcal{R} : \text{Sh}(X, \text{Ring})$
$D^b(A)$	Bounded derived objects of an abelian category A
$D^b(X, k)$	Bounded derived objects of $\text{Sh}(X, \text{Vect}_k)$, where $X : \text{Site}$ and $k : \text{Field}$
$X \rightarrow Y$	Morphisms from $X : T$ to $Y : T$

1.1.2 Global Notations. The following terms will be used throughout the write-up

$(-)^{\text{an}} : \text{Curv}_{\mathbb{C}}^{\text{sm}} \rightarrow \text{Riem}$	The complex analytification functor
$[n] : D^b(X, k) \rightarrow D^b(X, k)$	Cohomological shift by $n : \text{Int}$, for $X : \text{Site}$
$\Gamma(X, -) : D^b(X, k) \rightarrow D^b(\text{Vect}_k)$	Derived global sections for $X : \text{Site}$
$\mathcal{H}^i : D^b(X, k) \rightarrow \text{Sh}(X, \text{Vect}_k)$	The i -th cohomology sheaf, for $i : \text{Int}$ and $X : \text{Site}$
$H^i(X, -) : D^b(X, k) \rightarrow \text{Vect}_k$	The i -th hypercohomology, for $i : \text{Int}$ and $X : \text{Site}$
$\underline{V}_X : \text{Sh}(X, \text{Vect}_k)$	The constant sheaf associated with $V : \text{Vect}_k$ on $X : \text{Site}$
$f_* : D^b(X, k) \rightarrow D^b(Y, k)$	Derived pushforward along $f : X \rightarrow Y$ where $X, Y : \text{Site}$
$f^* : D^b(Y, k) \rightarrow D^b(X, k)$	Derived pullback along $f : X \rightarrow Y$ where $X, Y : \text{Site}$
$\otimes_k : D^b(X, k) \times D^b(X, k) \rightarrow D^b(X, k)$	Derived tensor product for $X : \text{Site}$
$\mathcal{H}om : D^b(X, k)^{\text{op}} \times D^b(X, k) \rightarrow D^b(X, k)$	Derived sheaf Hom for $X : \text{Site}$
$\mathcal{E}nd : D^b(X, k) \rightarrow D^b(X, k)$	Derived sheaf End for $X : \text{Site}$

where the derived functors are assumed to preserve boundedness.

1.2 Motivations

1.2.1. Monodromy groups are interesting mathematical objects that arise naturally in both geometric and arithmetic contexts. Controlling and understanding monodromy groups has been a central topic in both of these worlds.

1.2.2. On the geometric side,

1.2.3 A Question in Complex Geometry. Here is a

1.2.4. On the arithmetic side,

1.2.5 A Question in Number Theory. Here is a

1.3 Structure of the Write-up

1.3.1. In § 2

1.3.2. In § 3

1.3.3. In § 4

1.3.4. In § 5

1.4 Acknowledgements

1.4.1. The author thanks his advisor Prof. Chris Hall for his tremendous help.

1.4.2 Declaration of AI Use.

2 Local Systems and Lisse Sheaves

2.0.1 Notation. (Complex Geometric Setting) Define the following collection of notations

$X : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	A smooth projective connected curve over $\mathbb{C}$
$S : \text{Set}^{\text{fin}}$	A finite nonempty subset of closed points of X
$U : \text{Curv}_{\mathbb{C}}^{\text{sm}}$	The open subvariety $X \setminus S$
$j : U \rightarrow X$	The inclusion map
$m : \text{Nat}$	The cardinality of S

3 Perverse Sheaves and the Six Functor Formalism

4 Middle Convolution and Monodromy Groups

5 Mod- ℓ Monodromy of Hyperelliptic Curves